Verification checklist — extracted values

Printed from the files as they stood at 2026-10-15T10:09:35.164520Z (`braak2011_stage_counts.csv` sha256 df249382f0c7… · `meta_extraction.csv` sha256 2180448ab03a…). Every value below was extracted with Claude's help and is marked `verified: false`.

How this works. Check each value against the source and tick its box here. Then put an **x** in the `tick` column of the matching tick sheet (same ID) and run `python -m src.extraction.verification apply data/extracted/verification/ticks_<file>.csv` to see what would change, then the same command with `--apply`. Only rows you tick become `verified: true`. If a row has changed since this sheet was printed, the script refuses and asks for a fresh checklist.

1. Braak et al. 2011 — tau stage counts by age band

File: `data/extracted/braak2011_stage_counts.csv` (120 rows) · **Source:** Supplemental Digital Content 1 (Supplementary Table 1), `http://links.lww.com/NEN/A273` · **Used by:** Module 2 (all fits)

Open Supplementary Table 1 next to this sheet. For each cell, check that the number printed here equals the number in the supplement (same age row, same stage column), then tick its box. The grey code under each box (B-…) is the row ID on the tick sheet `ticks_braak2011_stage_counts.csv`. Row and column totals are computed from the file, so you can also compare them with the supplement's totals. Abstract totals for reference: 2,332 brains; 10 without AT8 tau; 58 at stages a–c; 38 at 1a; 236 at 1b.

Age (printed)	No changes	a	b	c	1a	1b	I	II	III	IV	V	VI	Total
00-10	**6** ☐ B-001	**2** ☐ B-002	**0** ☐ B-003	**0** ☐ B-004	**0** ☐ B-005	**0** ☐ B-006	**0** ☐ B-007	**0** ☐ B-008	**0** ☐ B-009	**0** ☐ B-010	**0** ☐ B-011	**0** ☐ B-012	8
11-20	**2** ☐ B-013	**8** ☐ B-014	**6** ☐ B-015	**2** ☐ B-016	**2** ☐ B-017	**2** ☐ B-018	**2** ☐ B-019	**0** ☐ B-020	**0** ☐ B-021	**0** ☐ B-022	**0** ☐ B-023	**0** ☐ B-024	24
21-30	**2** ☐ B-025	**17** ☐ B-026	**3** ☐ B-027	**1** ☐ B-028	**8** ☐ B-029	**15** ☐ B-030	**12** ☐ B-031	**3** ☐ B-032	**0** ☐ B-033	**0** ☐ B-034	**0** ☐ B-035	**0** ☐ B-036	61
31-40	**0** ☐ B-037	**9** ☐ B-038	**1** ☐ B-039	**1** ☐ B-040	**13** ☐ B-041	**38** ☐ B-042	**32** ☐ B-043	**2** ☐ B-044	**2** ☐ B-045	**1** ☐ B-046	**1** ☐ B-047	**0** ☐ B-048	100
41-50	**0** ☐ B-049	**4** ☐ B-050	**2** ☐ B-051	**1** ☐ B-052	**12** ☐ B-053	**78** ☐ B-054	**71** ☐ B-055	**13** ☐ B-056	**3** ☐ B-057	**2** ☐ B-058	**1** ☐ B-059	**1** ☐ B-060	188
51-60	**0** ☐ B-061	**0** ☐ B-062	**0** ☐ B-063	**1** ☐ B-064	**1** ☐ B-065	**60** ☐ B-066	**171** ☐ B-067	**63** ☐ B-068	**22** ☐ B-069	**5** ☐ B-070	**3** ☐ B-071	**4** ☐ B-072	330

Age (printed)	No changes	a	b	c	1a	1b	I	II	III	IV	V	VI	Total
61-70	**0** ☐ B-073	**0** ☐ B-074	**0** ☐ B-075	**0** ☐ B-076	**1** ☐ B-077	**33** ☐ B-078	**178** ☐ B-079	**165** ☐ B-080	**82** ☐ B-081	**16** ☐ B-082	**8** ☐ B-083	**9** ☐ B-084	492
71-80	**0** ☐ B-085	**0** ☐ B-086	**0** ☐ B-087	**0** ☐ B-088	**1** ☐ B-089	**9** ☐ B-090	**97** ☐ B-091	**156** ☐ B-092	**197** ☐ B-093	**54** ☐ B-094	**35** ☐ B-095	**21** ☐ B-096	570
81-90	**0** ☐ B-097	**0** ☐ B-098	**0** ☐ B-099	**0** ☐ B-100	**0** ☐ B-101	**1** ☐ B-102	**24** ☐ B-103	**87** ☐ B-104	**163** ☐ B-105	**91** ☐ B-106	**78** ☐ B-107	**40** ☐ B-108	484
91-100	**0** ☐ B-109	**0** ☐ B-110	**0** ☐ B-111	**0** ☐ B-112	**0** ☐ B-113	**0** ☐ B-114	**2** ☐ B-115	**8** ☐ B-116	**22** ☐ B-117	**22** ☐ B-118	**12** ☐ B-119	**9** ☐ B-120	75
Total	10	40	12	6	38	236	589	497	491	191	138	84	2332

Checked by: _____ Date: _____ Notes: _____

2. Meta-analysis extraction

File: `data/extracted/meta_extraction.csv` (37 rows) · **Used by:** Module 5. Citations and licences: `data/extracted/meta_extraction.sources.md`

For each row, open the paper at the place given (PMCID and locator), compare the numbers with the paper, and tick the big box only if all of them match. The small boxes are for your own working. The row ID (M-…) is the line on the tick sheet `ticks_meta_extraction.csv`. Excluded rows are listed too: their numbers are kept in the file, so they can be verified like the others.

☐ **M-01** · Beaman2023 · source `PMC9976965` · Results, section 'Highly BBB-permeable beta-blockers are associated with reduced risk of Alzheimer's disease', paragraph 2 (refers to Table 2/Fig. 3); Table 2 itself reports only absolute risks and risk differences

Exposure	lipophilic_beta_blocker (high BBB-permeability beta-blockers (carvedilol, propranolol))
Comparator	active: low BBB-permeability beta-blockers (atenolol, bisoprolol; sotalol excluded from primary cohort)
Outcome	Alzheimer's disease (10-year standardized risk; death as competing risk)
Number to check	**RR 0.76 (95% CI 0.61–0.95)**
Verbatim in file	*The 10-year standardized absolute risk of Alzheimer's disease was reduced by 24% in the high versus low BBB permeability group, with a risk ratio of 0.76 (CI: 0.61 to 0.95, P < 0.040; Table 2, Fig. 3).*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-02** · Desai2025 · source `PMC13127832` · Results, 'Comparative risk of ADRD', paragraph 2 (pooled HR, Analysis 1 ('as-treated' follow-up, censor at discontinuation/switch + 90-day grace)); database-specific HRs only in Figures 2-3 (not extracted)

Exposure	lipophilic_beta_blocker (highly brain-penetrant beta-blockers (propranolol, carvedilol))
Comparator	active: poorly brain-penetrant beta-blockers (atenolol, bisoprolol, sotalol)
Outcome	incident Alzheimer's disease and related dementias (ADRD)
Number to check	**HR 1.11 (95% CI 1.07–1.15)**
Verbatim in file	*For highly versus poorly brain-penetrant beta-blockers, the estimates indicated small differences that were statistically significant due to a very large sample size (pooled HR [95% CI] across Medicare and CPRD: 1.11 [1.07–1.15] in Analysis 1, 1.05 [1.02–1.08] in Analysis 2, 1.11 [1.07–1.15] Analysis 3, 1.05 [0.99–1.11] in Analysis 4)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-03** · Chen2025 · source `PMC12693173` · Results 3.2, paragraph 1 (text; Figure 2A). Terazosin/alfuzosin vs tamsulosin dementia HRs are only shown in Figure 2 (image) and were NOT extracted

Exposure	alpha1_blocker (doxazosin)
Comparator	active: tamsulosin (reference drug)
Outcome	dementia (ICD-10-CM F01-F03), 10-year window
Number to check	**HR 0.870 (95% CI 0.807–0.939)**
Verbatim in file	*Statistically significant HRs (HRs [95% CI]) for dementia were observed at 1 year (0.729 [0.615–0.865]), 3 years (0.769 [0.688–0.860]), and 10 years (0.870 [0.807–0.939])*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-04** · Fung2024 · source `PMC11340942` · Table 2, block 'Doxazosin | Tamsulosin', row '-overall', column 'Alzheimer's disease HR (95% CI)' (cells: N(%), HR for death, N(%), HR for PD, N(%), HR for AD)

Exposure	alpha1_blocker (doxazosin)
Comparator	active: tamsulosin (reference)
Outcome	Alzheimer's disease (CMS Chronic Conditions Data Warehouse indicator)
Number to check	**HR 0.84 (95% CI 0.76–0.92)**
Verbatim in file	*-overall | | 42,060(3.7) | 0.94(0.91,0.97) | 41,501(3.8) | 0.79(0.72,0.88) | 41,576(3.7) | 0.84(0.76,0.92)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-05** · Zhang2026 · source `PMC13644072` · Results (Merative Marketscan analysis), paragraph 'The association between a lower hazard of AD and the use of TZ/DZ/AZ persisted after adjustment'

Exposure	alpha1_blocker (terazosin, doxazosin or alfuzosin (TZ/DZ/AZ))
Comparator	active: tamsulosin
Outcome	Alzheimer's disease (broad ICD-9/10 code set incl. G30, F01, F03, G31.x, R41.x; first observed date)
Number to check	**HR 0.84 (95% CI 0.84–0.84)**
Verbatim in file	*The Cox proportional hazards model shows a lower hazard of AD in new users of TZ/DZ/AZ compared to new users of tamsulosin (HR = 0.84, t = -867.9, p < 10− 16, 95% CI: [0.84–0.84])*

Excluded from the analysis: E2: CI printed as [0.84-0.84] (t = -867.9) on an IPTW pseudo-population; no usable standard error

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-06** · Weber2023 · source `PMC9981768` · Results, paragraph 'An analysis of patient databases...' (Fig. 5)

Exposure	alpha1_blocker (terazosin, alfuzosin or doxazosin (TZ/DZ/AZ))
Comparator	active: tamsulosin
Outcome	dementia diagnosis (ICD-9/10 codes listed in Methods) in men with Parkinson's disease, up to 8 years
Number to check	**HR 0.75 (95% CI 0.57–0.98)**
Verbatim in file	*After matching for demographics, comorbidities, and disease duration, 1508 men (754 matched pairs) remained. This matching successfully reduced imbalance between the groups (Supplementary Table 1) and resulted in a statistically significant 25% reduction (hazard ratio (HR) = 0.75; 95% CI: 0.57, 0.98) in the hazard of being diagnosed with dementia*

Excluded from the analysis: E1: men with Parkinson's disease; outcome is dementia within PD

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-07** · Latvala2022 · source `PMC9542191` · Table 3, section 'α1-adrenoceptor antagonist use before lag time', row 'Any', column 'Adjusted OR (95% CI) b' (last cell)

Exposure	alpha1_blocker (any alpha1-adrenoceptor antagonist (tamsulosin and/or alfuzosin) use >=3 years before index (before lag time))
Comparator	non_user: no use of alpha1-adrenoceptor antagonists
Outcome	Alzheimer's disease (clinically verified, NINCDS-ADRDA/DSM-IV)
Number to check	**OR 1.10 (95% CI 1.07–1.14)**
Verbatim in file	*Any | 7938 (32.3) | 27 515 (28.0) | 1.24 (1.20–1.27) | 1.16 (1.12–1.19) | 1.10 (1.07–1.14)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-08** · GarciaArgibay2022 · source `PMC9857015` · Table 2, outcome block 'all-cause dementia', row 'Adjusted', column 'α-Blockers HR (95% CI)' (5th cell)

Exposure	alpha1_blocker (α-blockers (alone; agents not specified in main text))
Comparator	non_user: individuals unexposed to any of the study drugs (5-ARIs or α-blockers) at time t
Outcome	all-cause dementia
Number to check	**HR 0.71 (95% CI 0.68–0.73)**
Verbatim in file	*Adjusteda | 1.22 (1.17-1.28) | <.001 | 1.10 (1.01-1.20) | .04 | 0.71 (0.68-0.73) | <.001 | 0.73 (0.70-0.77) | <.001*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-09** · Colbourne2022 · source `PMC9708561` · Table 2 (panel A: no prior neuropsychiatric diagnosis), row 'Dementia', column 'Risk ratio (95% CI)' (3rd numeric cell)

Exposure	dhp_calcium_channel_blocker (brain-penetrant dihydropyridine CCBs (felodipine, isradipine, nicardipine, nifedipine, nimodipine, nisoldipine))
Comparator	active: amlodipine (non-penetrant dihydropyridine) [NB: not one of the 4 priority contrasts; brain-penetrance contrast within DHP class]
Outcome	dementia (ICD-10 F01-F03, G30, G31.0, G31.2, G31.83) during 2-year exposure
Number to check	**RR 0.82 (95% CI 0.72–0.93)**
Verbatim in file	*Dementia | 1.0 | 1.2 | 0.82 (0.72–0.93) | 3.2 | 3.3 | 0.98 (0.87–1.10)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-10** · Wu2026 · source `PMC13572180` · Table 2, group 'Neuromodulators and adrenergic agents', row 'Propranolol | Supplements', column 'Risk ratio (95% CI)'

Exposure	lipophilic_beta_blocker (propranolol (initiation))
Comparator	active: initiators of supplements (active control group, e.g. vitamin B12, folic acid; designed as minimal-effect control) [NB: not a priority contrast]
Outcome	ADRD (AD, vascular, Lewy body, frontotemporal dementia) after MCI diagnosis, up to 5 years
Number to check	**RR 0.56 (95% CI 0.37–0.85)**
Verbatim in file	*Propranolol | Supplements | 0.56 (0.37–0.85)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-11** · Fang2022 · source `PMC8751379` · Results text (population-based validation paragraph; Figs. 5-6). Nadolol and sotalol HRs only in Figs. 5-6 (not extracted)

Exposure	hydrophilic_beta_blocker (atenolol)
Comparator	non_user: matched atenolol non-users (control population)
Outcome	Alzheimer's disease
Number to check	**HR 0.949 (95% CI 0.923–0.976)**
Verbatim in file	*and atenolol (HR = 0.949, 95% CI 0.923–0.976, P = 2.8 × 10−4) are associated with a reduced risk of AD compared with matched control populations*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-12** · Song2023 · source `PMC10191296` · Table 2, row 'Vascular endothelial growth factor A (other) | Carvedilol', column 'AD only N = 8,332' (2nd OR cell)

Exposure	lipophilic_beta_blocker (carvedilol)
Comparator	non_user: beneficiaries who did not use carvedilol
Outcome	Alzheimer's disease (AD only case group)
Number to check	**OR 0.90 (95% CI 0.81–0.99)**
Verbatim in file	*Vascular endothelial growth factor A (other)b | Carvedilolb | 0.81 (0.77, 0.86) | 0.90 (0.81, 0.99) | 0.78 (0.59, 1.01) | 0.72 (0.62, 0.82) | 0.80*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-13** · Kern2019 · source `PMC6909196` · Table 1 ('Results from the pooled meta-analyses across four US claims databases'), row 'Atomoxetine', columns 'IRR', 'Lower 95%', 'Upper 95%'

Exposure	atomoxetine (atomoxetine)
Comparator	non_user: same individuals' unexposed period before atomoxetine initiation (within-person)
Outcome	incident dementia (>=2 dementia claims within 12 months; includes AD, vascular dementia, others)
Number to check	**IRR 0.69 (95% CI 0.39–1.20)**
Verbatim in file	*Atomoxetine | 407,883 | 292 ± 308 | 80 | 112 | 0.69 | 0.39 | 1.20 | .189 | 0.68*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-14** · Then2017 · source `PMC5383251` · Table 3 ('Risk of dementia with the use of antidepressants from different classes (reference = control)'), row 'Monoamine oxidase inhibitors (MAOI)', column 'Multivariate HR [95% CI]'

Exposure	mao_a_inhibitor (MAOI (moclobemide))
Comparator	non_user: matched controls with no antidepressant use
Outcome	dementia (ICD-9-CM 290, 294.1) or Alzheimer's disease (331.0)
Number to check	**HR 4.94 (95% CI 2.17–11.24)**
Verbatim in file	*Monoamine oxidase inhibitors (MAOI) | 155 | (2.66%) | 6.72 [2.97–15.13] | 4.94 [2.17–11.24]***

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-15** · Ding2025 · source `PMC11772729` · Table 2, block 'C07', row 'Using ≥ 10 years', column 'Odds ratio (95% CI) Model 2' (last cell)

Exposure	beta_blocker_any (beta-blockers (ATC C07), use >=10 years)
Comparator	non_user: non-users of beta-blockers
Outcome	dementia (ICD-10 F00, F01, F02, F03, F05, G30)
Number to check	**OR 0.82 (95% CI 0.79–0.85)**
Verbatim in file	*Using ≥ 10 years* | *49,026 (5.6)* | *4432 (5.0)* | *0.91 (0.88–0.94)* | *0.82 (0.79–0.85)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-16** · Wandell2018 · source `PMC6690390` · Table 3, row 'All beta blockers', column 'Model 4 HR (95% CI)' (last cell)

Exposure	beta_blocker_any (all beta blockers (C07))
Comparator	non_user: AF patients without beta-blocker prescription
Outcome	incident dementia (first hospital diagnosis)
Number to check	**HR 0.92 (95% CI 0.77–1.09)**
Verbatim in file	*All beta blockers* | *0.90 (0.77–1.04)* | *0.81 (0.69–0.95)* | *0.89 (0.75–1.06)* | *0.92 (0.77–1.09)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-17** · Wandell2018 · source `PMC6690390` · Table 3, row 'Vessel active calcium channel blockers', column 'Model 4 HR (95% CI)' (last cell)

Exposure	dhp_calcium_channel_blocker (vessel active calcium channel blockers (Methods: C08CA and C08DB; Table 2: C08C, C09DB))
Comparator	non_user: AF patients without vessel-active CCB prescription
Outcome	incident dementia (first hospital diagnosis)
Number to check	**HR 0.98 (95% CI 0.80–1.20)**
Verbatim in file	*Vessel active calcium channel blockers* | *0.90 (0.75–1.07)* | *0.90 (0.75–1.09)* | *0.98 (0.81–1.19)* | *0.98 (0.80–1.20)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-18** · TonegawaKuji2026 · source `PMC12746425` · Supporting Information file TRC2-12-e70193-s002.xlsx, sheet 'Table S14' (Results of trials comparing candidate vs. active comparators in the over-70 cohort), spreadsheet row 13 (No. 9, zolpidem vs suvorexant), columns 27-29 'IP-weighted Kaplan-Meier estimate: Hazard Ratio / 95%CI lower / 95% CI upper' (P value col 30 = 0.579561287)

Exposure	dual_orexin_antagonist (zolpidem (Z-drug; exposure group))
Comparator	active: suvorexant (DORA; reference group)
Outcome	first Alzheimer's disease dementia diagnosis (incidence) within 3 years
Number to check	**HR 1.173683544 (95% CI 0.666042539–2.068235854) · invert_ratio: true (the code flips the printed ratio; check the printed numbers as printed)**
Verbatim in file	*9 | zolpidem | suvorexant | 0.043075117 | 0.026509573 | 0.052762372 | 0.047784174 | 0.004978199 | 0.0131189639155914 | 0.0125818760074311 | 0.0136560518237516 | 0.000631066 | -0.053496197 | 0.034900399 | 0.243512974 | 1.104180912 | 1.33138254266443 | 1.31788469196951 | 1.34501863912128 | 1.108085065 | 0.49108717 | 2.84930207 | 0.265469062 | 1.332595917 | 0.829983892 | 2.139573905 | 1.173683544 | 0.666042539 | 2.068235854 | 0.579561287 | non_significant*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-19** · Murray2018 · source `PMC5880772` · Electronic supplementary material ESM 1 (11606_2017_4281_MOESM1_ESM.docx), Supplementary Table S1, row 'Taking a beta-blocker', columns 'HR' and '95% CI' (reference row 'Not prescribed antihypertensive medication')

Exposure	beta_blocker_any (beta-blocker (as labelled in Supplementary Table S1; 'beta-adrenergic blockers' in Methods))
Comparator	non_user: not prescribed antihypertensive medication (untreated hypertensive participants; reference)
Outcome	incident dementia (two-stage clinical diagnosis)
Number to check	**HR 0.43 (95% CI 0.26–0.74)**
Verbatim in file	*Taking a beta-blocker | 0.43 | 0.26-0.74 | 0.002*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-20** · Murray2018 · source `PMC5880772` · Electronic supplementary material ESM 1 (11606_2017_4281_MOESM1_ESM.docx), Supplementary Table S1, row 'Taking an alpha-beta-blocker', columns 'HR' and '95% CI' (reference row 'Not prescribed antihypertensive medication')

Exposure	alpha_beta_blocker (alpha-beta-blocker (agents not specified; class includes e.g. carvedilol/labetalol))
Comparator	non_user: not prescribed antihypertensive medication (untreated hypertensive participants; reference)
Outcome	incident dementia (two-stage clinical diagnosis)
Number to check	**HR 0.83 (95% CI 0.28–2.46)**
Verbatim in file	*Taking an alpha-beta-blocker | 0.83 | 0.28-2.46 | 0.73*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-21** · Murray2018 · source `PMC5880772` · Electronic supplementary material ESM 1 (11606_2017_4281_MOESM1_ESM.docx), Supplementary Table S1, row 'Taking a central adrenergic agonist', columns 'HR' and '95% CI' (reference row 'Not prescribed antihypertensive medication')

Exposure	central_alpha2_agonist (centrally acting adrenergic agonist (agents not specified; class may include clonidine, methyldopa, guanfacine))
Comparator	non_user: not prescribed antihypertensive medication (untreated hypertensive participants; reference)
Outcome	incident dementia (two-stage clinical diagnosis)
Number to check	**HR 0.56 (95% CI 0.25–1.27)**
Verbatim in file	*Taking a central adrenergic agonist | 0.56 | 0.25-1.27 | 0.16*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-22** · Gray2015 · source `PMC4358759` · Table 3, Dementia block, row '>1095', column 'Adjusted HR | 95% CI' (last two cells)

Exposure	anticholinergic (10-year cumulative strong anticholinergic use, total standardized daily doses (TSDD), time-varying; most recent 12 months excluded)
Comparator	non_user: 0 TSDD (non-use)
Outcome	incident dementia (research-based criteria; multidisciplinary consensus conference after CASI screening)
Number to check	**HR 1.54 (95% CI 1.21–1.96)**
Verbatim in file	*>1095 | 4022 | 184 | 1.77 | 1.40-2.23 | 1.54 | 1.21-1.96*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-23** · Richardson2018 · source `PMC5915701` · Table 2, 'Number of DDDs (ACB score)' section, subheading '3:', row '>1460', last column 'Adjusted at end of DEP'

Exposure	anticholinergic (number of defined daily doses (DDDs) of drugs with ACB score 3 prescribed during the drug exposure period (4-20 y before index): >1460 DDDs)
Comparator	non_user: 0 DDDs of ACB-3 drugs
Outcome	incident dementia (Read code diagnosis or cognitive enhancer with dementia code)
Number to check	**OR 1.31 (95% CI 1.22–1.41)**
Verbatim in file	*>1460 | 1052 (2.6) | 5077 (1.8) | | 1.59§ (1.49 to 1.71) | 1.40§ (1.30 to 1.50) | 1.31§ (1.22 to 1.41)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-24** · Coupland2019 · source `PMC6593623` · Table 4, 'Exposure in the 1 to 11 Years Before Index Date', row '>1095', column 'OR (95% CI) Adjusted'

Exposure	anticholinergic (total cumulative anticholinergic drug exposure (56 strong anticholinergics; TSDDs) in the 1 to 11 years before index date: >1095 TSDDs)
Comparator	non_user: nonuse
Outcome	incident dementia (clinical code or ONS death record, or anti-dementia drug prescription)
Number to check	**OR 1.49 (95% CI 1.44–1.54)**
Verbatim in file	*>1095 | 9800 (16.7) | 27 436 (12.2) | 1.58 (1.53-1.62) | 1.49 (1.44-1.54)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-25** · Grossi2019 · source `PMC6802337` · Table 2, 'ACB3 use' block, row 'Recurrent', column 'Adjusted IRR (95% CI)'

Exposure	anticholinergic (recurrent use of ACB score 3 anticholinergics (self-reported use at both Y0 and Y2))
Comparator	non_user: no ACB3 use at Y0 or Y2 ('None')
Outcome	incident dementia by Y10 (study diagnosis, AGECAT score >=3)
Number to check	**IRR 1.68 (95% CI 1.00–2.82)**
Verbatim in file	*Recurrent | 12 | 70 | 18.6 | 2.05 (1.18,3.56) | 1.68 (1.00,2.82)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-26** · Suh2020 · source `PMC7391507` · Table 4, '2-10 years before index year' block, row 'High (≥2)', column 'KABS aOR (95% CI)'

Exposure	anticholinergic (average daily anticholinergic burden score over 2-10 years before index year by Korean Anticholinergic Burden Scale (KABS, incl. score-1 drugs): High (>=2))
Comparator	non_user: Minimal (<0.25)
Outcome	incident dementia (claims-based)
Number to check	**OR 1.71 (95% CI 1.55–1.87)**
Verbatim in file	*High (≥2) | 1.71 (1.55–1.87) | 1.22 (1.12–1.33) | 1.60 (1.38–1.86) | 1.41 (1.14–1.75)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-27** · Joung2019 · source `PMC6494800` · Table 3, 'All subjects (n = 191,805)', row '≥120', last column 'Model II'

Exposure	anticholinergic (average annual prescribed amount of strong anticholinergics (Beers strong or ACB 2-3), standardized doses/year: >=120)
Comparator	non_user: 0-9 doses/year (least exposed)
Outcome	incident Alzheimer's disease (ICD-10 F00/G30 plus anti-AD drug)
Number to check	**HR 1.39 (95% CI 1.30–1.50)**
Verbatim in file	*≥120 | 1,053 (12.1) | 3.87 (3.63–4.10) | 1.98 (1.86–2.12) | 1.57 (1.47–1.68) | 1.39 (1.30–1.50)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-28** · Yang2017 · source `PMC5383287` · Table 3, row 'Oxybutynin', column 'After match HR(95%CI)'

Exposure	anticholinergic (oxybutynin use (patients with DM))
Comparator	non_user: matched control DM patients not receiving oxybutynin, solifenacin or tolterodine
Outcome	incident dementia (claims-based)
Number to check	**HR 2.35 (95% CI 1.96–2.81)**
Verbatim in file	*Oxybutynin | 2.68(2.47–2.91) | <0.001 | 2.35(1.96–2.81) | <0.001*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-29** · Poonawalla2023 · source `PMC10311860` · Table 2, 'Anticholinergic polypharmacy' block, row '≥ 4 ACH meds', column 'Adjusted Hazard Ratio (95% CI)'

Exposure	anticholinergic (time-varying anticholinergic (ACB-listed) polypharmacy: >=4 ACH medications on a given day)
Comparator	non_user: no ACH exposure (person-time)
Outcome	incident dementia/Alzheimer's disease (combined, claims-based)
Number to check	**HR 2.59 (95% CI 1.06–6.32)**
Verbatim in file	*≥ 4 ACH meds | 6 | 78 | 76.92 | 2.59 (1.06—6.32)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-30** · Zhu2025 · source `PMC12538848` · Table 2, '1-year lag time' > 'Strong anticholinergics', row '≥ 1095', column 'Model 3'

Exposure	anticholinergic (cumulative DDDs of strong anticholinergics (ACB score 2-3) prescribed >=1 year before dementia onset (1-year lag): >=1095 DDDs)
Comparator	non_user: 0 DDDs of strong anticholinergics
Outcome	incident all-cause dementia
Number to check	**OR 1.66 (95% CI 1.55–1.79)**
Verbatim in file	*≥ 1095 | 2131 (1.1) | 1176 (0.6) | 1.91 (1.78, 2.05)‡ | 1.73 (1.61, 1.87)‡ | 1.66 (1.55, 1.79)‡*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-31** · Iyen2024 · source `PMC11580265` · Table 4, 'Any anticholinergic drug treatment for overactive bladder', row 'Use of anticholinergic drugs for overactive bladder', column 'Adjusted odds ratio† (95% CI)'

Exposure	anticholinergic (use of any anticholinergic drug for overactive bladder 3-16 years before diagnosis/index date)
Comparator	non_user: no use of anticholinergic drugs to treat overactive bladder
Outcome	incident dementia (all subtypes)
Number to check	**OR 1.18 (95% CI 1.16–1.20)**
Verbatim in file	*Use of anticholinergic drugs for overactive bladder | 15 418 (9.03) | 63 369 (7.88) | 1.15 (1.13 to 1.17) | 1.18 (1.16 to 1.20)*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-32** · Chi2025 · source `PMC12695687` · Table 3, 'TSDD' block, row 'TSDD', column 'Model 2 HR (95% CI)'

Exposure	anticholinergic (any anticholinergic exposure in the year before baseline, TSDD > 0 (ACB-scale drugs))
Comparator	non_user: Non-TSDD (TSDD = 0)
Outcome	incident dementia
Number to check	**HR 1.36 (95% CI 0.93–1.99)**
Verbatim in file	*TSDD | 1.5 (1.08, 2.24) | 0.0189* | 1.36 (0.93, 1.99) | 0.1107*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-33** · Gildengers2023 · source `PMC9974875` · Table 4, 'Total ACR score' block, row '>3', column 'Model 5 HR (95% CI)'

Exposure	anticholinergic (total Anticholinergic Rating (ACR) score >3 (sum of ACR scores of all prescription and OTC drugs))
Comparator	non_user: total ACR score 0
Outcome	incident dementia (CDR >= 1)
Number to check	**HR 1.1 (95% CI 0.6–1.9)**
Verbatim in file	*>3 | | | 1.1 (0.6, 1.9) | 0.825 | |*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-34** · Liu2020 · source `PMC7033580` · Table 2, row 'With', columns 'aHR | 95% CI | 95% CI'

Exposure	anticholinergic (anticholinergic medication use (with))
Comparator	non_user: without anticholinergic medications (matched controls)
Outcome	incident dementia (ICD-9-CM)
Number to check	**HR 1.043 (95% CI 0.958–1.212)**
Verbatim in file	*With | 1.095 | 0.966 | 1.225 | 0.127 | 1.043 | 0.958 | 1.212 | 0.139 | |*

Excluded from the analysis: E2: CI asymmetric on the log scale as printed (e.g. 1.043 (0.958-1.212)); SE cannot be backed out

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-35** · PhamNguyen2025 · source `PMC12022375` · Table 2, 'Time-varying ACB categories', row '3+', column 'Adjusted HR (95% CI)'

Exposure	anticholinergic (time-varying total ACB score category 3+)
Comparator	non_user: total ACB 0
Outcome	incident Parkinson disease dementia (PDD)
Number to check	**HR 1.03 (95% CI 0.67–1.58)**
Verbatim in file	*3+ | 1.03 (0.67-1.58) | 0.9*

Excluded from the analysis: E1: Parkinson's disease population; outcome is Parkinson disease dementia

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-36** · Amrein2026 · source `PMC13570850` · Results > 'Long-Term Tricyclic Antidepressants Users', paragraph 1

Exposure	anticholinergic (long-term (>2 years) antidepressant use with at least one tricyclic antidepressant (TCA) prescription)
Comparator	non_user: propensity score-matched no-use (no antidepressant) control group
Outcome	incident dementia (diagnosis or antidementia drug)
Number to check	**OR 1.73 (95% CI 1.33–2.25)**
Verbatim in file	*Our analysis revealed significantly higher odds for developing dementia for individuals in the long-term tricyclic antidepressant use group compared with those in the no-use control group (OR 1.73, 95% CI 1.33–2.25, p < 0.001)*

Excluded from the analysis: E3: antidepressant (TCA) users vs non-users of antidepressants: confounded by indication (depression)

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

☐ **M-37** · Li2025 · source `PMC12274944` · Table 3, row 'Anticholinergic vs beta-3 agonist (reference)', column 'HR (95% CI)'

Exposure	anticholinergic (new use of an OAB anticholinergic (oxybutynin, solifenacin, tolterodine, trospium, darifenacin, propiverine, fesoterodine), adults 18-64 yr)
Comparator	active: new use of OAB beta-3 agonist (mirabegron) - ACTIVE comparator, not non-use
Outcome	incident dementia (validated administrative definition)
Number to check	**HR 0.99 (95% CI 0.86–1.15)**
Verbatim in file	*Anticholinergic vs beta-3 agonist (reference) | 0.99 (0.86–1.15) | 0.9*

☐ estimate ☐ lower CI ☐ upper CI ☐ measure (HR/RR/OR/IRR) ☐ exposure and comparator the right way round Notes: _____

Checked by: _____ Date: _____